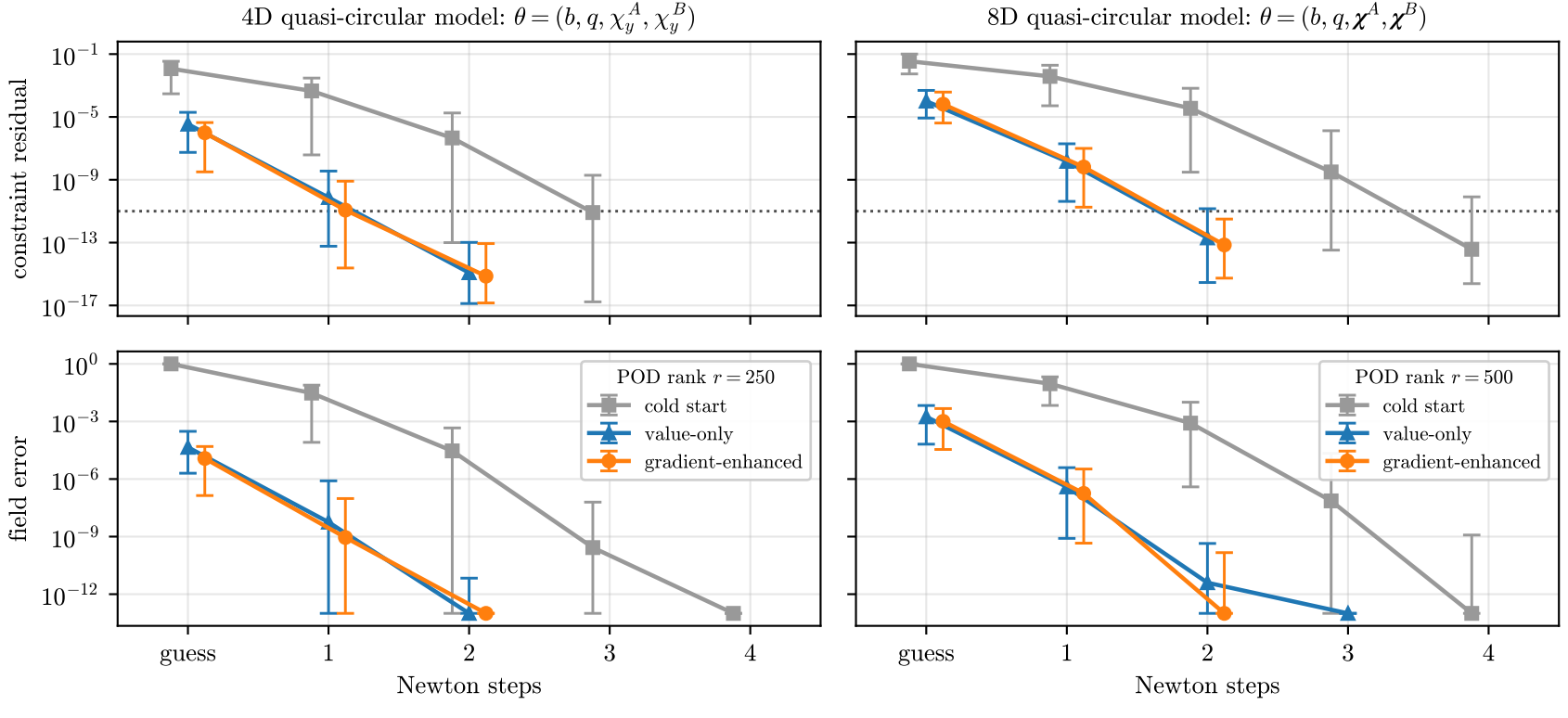
4D quasi-circular model: $\theta = (b, q, \chi_y^A, \chi_y^B)$
8D quasi-circular model: $\theta = (b, q, \boldsymbol{\chi}^A, \boldsymbol{\chi}^B)$
constraint residual
field error
10^{-1}
10^{-5}
10^{-9}
10^{-13}
10^{-17}
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
POD rank $r = 250$
POD rank $r = 500$
cold start
value-only
gradient-enhanced
guess
1
2
3
4
Newton steps